

Experiment 4

The Impact of Rigid Bodies

2022080054 Yong Rui Qian

1. Experiment Objectives

- Improve comprehension of the Impact section in theoretical mechanics.
- Become more acquainted with the methodology used in impact analysis.
- Achieve a thorough understanding of the mechanical meaning of restore coefficient and impulse ratio.
- Introduce a methodology to simplify and solve three-dimensional impact problems.

2. Experiment Contents

- Measure the restore coefficient
- Measure the impulse ratio
- The introduction of methodology to simplify a 3D impact problem

3. Experiment Principles

- Measurement of restore coefficient e

Let initial height h_0 , velocity of steel ball before impact v_1^* , max bounce height of ball h_{max} , velocity of mass center of ball V_1 . Get, $v_1 = \sqrt{2gh_0}$, $V_1 = -\sqrt{2gh_{max}}$ (negative indicates direction of velocity points to -Z direction. Hence, restore coefficient is defined as $e = -\frac{V_1 - V_A}{v_1 - v_A}$, and since velocities of block A before and after impact $V_A = 0$, then $e = \sqrt{\frac{h_{max}}{h_0}}$.

- Measurement of impulse ratio μ

The impulse ratio is defined as $\mu = \frac{|P_t|}{|P_n|}$. (ratio of tangential component to normal component)

After several matrix computations, we get the velocity of mass center of ball $\vec{V}_1$

$$\begin{Bmatrix} V_{1y} \\ V_{1z} \end{Bmatrix} = [A]^{-1} \begin{Bmatrix} V_{1t} \\ V_{1n} \end{Bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{Bmatrix} V_{1t} \\ V_{1n} \end{Bmatrix}$$

By conducting kinetic analysis of steel ball after impact, we get

In the direction of axis Y, $y = V_{1y}t$

In the direction of axis Z, $z = a + V_{1z}t - \frac{1}{2}gt^2$

Hence we get equation of locus, $z = a + y \frac{V_{1z}}{V_{1y}} - \frac{g}{2} \frac{1}{V_{1y}^2} y^2$

The equations above are solved simultaneously to get both tangential vector

$|P_t| = |m(V_{1t} - v_{1t})|$ and normal vector $|P_n| = |m(V_{1n} - v_{1n})|$.

Finally, by substituting these vectors into the expression above, we get $\mu = \left| \frac{(V_{1t} - v_{1t})}{(V_{1n} - v_{1n})} \right|$.

- Introduction to the methodology of three-dimensional impact simplification
The steel ball's descent, impact with blocks A and B, and landing on the floor involve five stages—three periods of uniformly accelerated movement and two impacts. The critical aspect is the analysis of the impact procedure.

The analysis employs natural coordinates, defining the normal direction as the impacted surface's normal vector, the tangential direction as the projection of the ball's velocity before impact, and the second normal direction as their cross product—forming the natural coordinates system, o-nt.

To simplify the 3D impact, assume impacts occur only in the o-nt plane, neglecting impulses in the second normal vector direction. Treating the steel ball as a rigid body, ignoring rotation, simplifies the problem. The revised definition of normal impulse uses the steel ball's velocity.

The impact procedure is studied in natural coordinates, while uniformly accelerated motion is described in global Cartesian coordinates. Coordinate transformation is necessary, converting velocities and locations from local to global coordinates.

Impact:

Assume that the velocity of the steel ball is $\vec{v} = \{v_n \quad v_t \quad v_b\}^T$ before a certain impact; And

$\vec{V} = \{V_n \quad V_t \quad V_b\}^T$ after the impact. Assume the velocity of the impacted surface stays zero.

In normal direction, $V_n = -ev_n$

In tangential direction, $V_t = \mu(1 + e)v_n + v_t$

Uniformly Accelerated Motion:

Assume the location of the *i*th impact is $\{X \quad Y \quad Z\}^T$, the speed of the ball after impact is

$\vec{V} = \{V_x \quad V_y \quad V_z\}^T$, the equations of the uniformly accelerated movement afterwards are,

$$\begin{cases} x = X + V_x t \\ y = Y + V_y t \\ z = Z + V_z t - 0.5gt^2 \end{cases}$$

The equation for the *i*+1th impacted surface is, $Ax + By + Cz + D = 0$

Equations above are solved simultaneously.

4. Experimental Instruments

Number	1	2	3	4	5	6	7	8
Instrument	Starter	Block B	Fence	Adjusting Screw (X3)	Floor	Track	Block A	Pillar

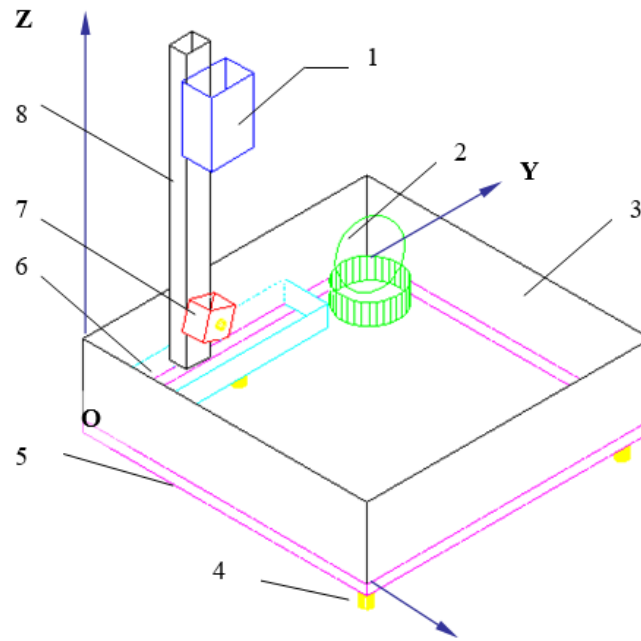


Figure 1 Experiment Facilities

5. Experiment Procedures

- (1) Record h_0 . Ensure block A is aligned horizontally. Press the button on the right side of the starter. The steel ball inside the starter will be released and fall freely due to gravity.
- (2) Read and record the data h_i from the programme. Repeat steps 1 and 2 until 6 valid data is recorded and calculate the average value $\bar{h}_{max}$.
- (3) Calculate restore coefficient $e = \sqrt{\frac{\bar{h}_{max}}{h_0}}$.
- (4) Adjust the angle of inclination of block A to 13° . Put a coordinate paper(or a blank paper) on the floor of the facility before releasing the ball and record the location of the paper. Then put a carbon paper on it and fasten the paper with a block. When the ball lands on the floor, it will leave a mark on the coordinate paper. The landing location of the ball is the sum of the location of the mark on the coordinate paper and the location of the paper in the global O-XYZ coordinates. Repeat this step for 6 times and calculate the average landing location of ball $\bar{b}$.
- (5) Calculate impulse ratio μ .
- (6) Install block B to the impact table (22°) and check if steel ball will bounce off both blocks. The following procedure is similar to step 4 (carbon paper for landing location). Measure and calculate the average landing location $\bar{x}$ and $\bar{y}$.
- (7) Calculate the margin of error and distance between theoretical landing location.
- (8) Repeat steps 4 and 5 with a different set of equipment.

6.Experiment Data

Equipment 5

$$h_0 = 270.0 \text{ (mm)}, \quad \bar{h}_{\max} = \left(\sum_{i=1}^6 h_i / 6 \right) = 233.4 \text{ (mm)}, \quad e = \sqrt{\frac{\bar{h}_{\max}}{h_0}} = 0.9298$$

$$\alpha = 13^\circ, \quad \beta = 22^\circ, \quad d = 30 \text{ (mm)}$$

$$\bar{b} = 400 - 30 - \left(\sum_{i=1}^6 b'_i / 6 \right) = 350.7 \text{ (mm)}, \quad \mu = 0.0431$$

Experimental landing position: $x = 236.8333 \text{ (mm)}$, $y = 38.3333 \text{ (mm)}$

1. Theoretical landing location: $x_T = 235.3 \text{ (mm)}$, $y_T = 38.4 \text{ (mm)}$

2. Absolute error: $\Delta x = 1.5333 \text{ (mm)}$, $\Delta y = 0.0667 \text{ (mm)}$

3. Distance between two landing location: $\sqrt{(\Delta x)^2 + (\Delta y)^2} = 1.5348 \text{ (mm)}$

Measurement of e	1	2	3	4	5	6
$h_i \text{ (mm)}$	224.4	233.9	233.9	238.2	236.1	233.9
Measurement of μ	1	2	3	4	5	6
$b'_i \text{ (mm)}$	17.5	21.0	16.0	19.5	27.5	14.5
Measure of landing location	1	2	3	4	5	6
$x \text{ (mm)}$	242.0	240.5	240.0	243.5	221.0	234.0
$y \text{ (mm)}$	26.5	21.5	47.5	59.0	50.0	25.5

Equipment 1

$$\alpha = 17^\circ, \quad \beta = 22^\circ, \quad d = 20 \text{ (mm)}$$

$$\bar{b} = 400 - d - \left(\sum_{i=1}^6 b'_i / 6 \right) = 332.4167 \text{ (mm)}, \quad \mu = 0.0652$$

Measurement of μ	1	2	3	4	5	6
$b'_i \text{ (mm)}$	46.5	48.0	49.0	48.5	46.5	47.0

7.Questions

- There are several equipment with different configurations and operation methods in the laboratory. Combining with the observation and discussion of the differences in the equipment in each experiment, analyze the reasons for the difference in the degree of dispersion between the different experimental groups. (write the most important 3 reasons)
- Analyze the origin of the error between the theoretical and experimental landing point of the ball in the 3D impact test based on your observation. (write the most important 3 reasons)

i) The degree of dispersion is influenced by the method release. The release of steel ball from a magnetic switch provides a controlled and reproducible release mechanism, minimizing variations in the initial conditions of the experiment. In contrast, mechanical starter may introduce variations in the initial velocity of steel ball. It might even cause steel ball to spin, hence gaining angular momentum. The scenarios mentioned above all contributes to the degree of dispersion.

Second of all, the degree of flatness and smoothness of surface of impact on Block A can also influence the degree of dispersion of steel ball landing location. Lastly, the degree of roundness of steel ball could also affect the end results.

ii) The data we obtained to calculate the restore coefficient might not be accurate due manual operational errors. Thus, the calculated coefficient could be far away from the actual coefficient. We then further the range of error as we use it to calculate the impulse ratio. Eventually, we use both restore coefficient and impulse ratio to calculate the theoretical landing coordinates.

Secondly, the equipments in laboratory might not be in peak form and shape and might have to be calibrated being using. Differences in sensor accuracy and alignment may impact accuracy of measurements.

Lastly, theoretical models may make assumptions that do not perfectly mirror real-world conditions. Deviations from these assumptions can contribute to discrepancies between theory and experiment.

8. Original Data Table

附体碰撞实验

姓名	班级	学号	教师签字: 	组号: 05 / 01
杨志峰	物理24	20220300424		
—	—	—	2023 年 12 月 13 日	

组号: 5

测量恢复系数	1	2	3	4	5	6
h_i (mm)	184.6 224.4	223.9	233.9	232.9	238.2	236.1
$h_0 = 370$ (mm)	$\bar{h}_{\max} = \left(\sum_{i=1}^6 h_i / 6 \right) = 233.4$ (mm)			$e = \sqrt{\frac{\bar{h}_{\max}}{h_0}} = 0.9278$		保留 4 位有效数字
测量冲量比/次数	1	2	3	4	5	6
b_i' (mm)	17.1	21.0	16.0	19.5	23.5	14.5
$\alpha = 13^\circ$	$\bar{b} = 400 - d - \left(\sum_{i=1}^6 b_i' / 6 \right) = \frac{310.6667}{\cancel{14.4222}} \text{ (mm)}$			冲量比 $\mu = 0.0431$		保留 3 位有效数字
三维碰撞落点位置: $\alpha =$ 与测冲量比时相同 $\beta = 22^\circ$						
	1	2	3	4	5	6
x (mm)	242.0	240.4	240.0	243.5 240.0	221.0	234.8
y (mm)	26.5	21.5	47.5 49.5	59.0	50.0	25.5
测量值平均 x	236.833 mm	计算值 x		绝对误差 x	1.5333 mm	与理论落点的距离
测量值平均 y	32.333 mm	计算值 y		绝对误差 y	0.0667 mm	1.5398 mm

组号: 5 /

恢复系数: ~~0.9278~~ 0.9634

测量冲量比/次数	1	2	3	4	5	6
b_i' (mm)	46.5	48.0	49.0	48.5	46.5	47.0
$\alpha = 13.7^\circ$	$\bar{b} = 400 - d - \left(\sum_{i=1}^6 b_i' / 6 \right) = 332.9167 \text{ (mm)}$			冲量比 $\mu = 0.0632$		保留 3 位有效数字

$d = 20$

$\bar{b}_i' = 47.5333$